

Problem Set #3

Q1) Panel Data

(a) Suppose the researcher runs the following regression

$$y = X\beta + u$$

where $u = \alpha + \epsilon$

$$\alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}_{(n \times T) \times 1}$$

$$\hat{\beta}^{OLS} = (X'X)^{-1} X'y$$

Substituting for the true y

$$\hat{\beta}^{OLS} = (X'X)^{-1} X'(X\beta + \alpha + \epsilon)$$

$$\hat{\beta}^{OLS} = (X'X)^{-1} X'X \beta + (X'X)^{-1} X' \alpha + (X'X)^{-1} X' \epsilon$$

$$\hat{\beta}^{OLS} = \beta + \left(\frac{1}{nT} X'X \right)^{-1} \frac{1}{nT} X' \alpha + \left(\frac{1}{nT} X'X \right)^{-1} \frac{1}{nT} X' \epsilon$$

By LLN:

$$\text{plim}_{n \rightarrow \infty} \left(\frac{1}{nT} X'X \right)^{-1} \rightarrow S_{X'X}^{-1}$$

$$\text{plim}_{n \rightarrow \infty} \frac{1}{nT} X' \epsilon \rightarrow 0$$

$$\frac{1}{nT} X' \alpha = \begin{bmatrix} X_{111} & X_{121} & X_{131} & \dots & X_{1T1} & X_{211} & \dots & X_{nT1} \\ X_{112} & X_{122} & \dots & \dots & \dots & \dots & \dots & X_{nT2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & X_{nTk} \end{bmatrix} \times \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{nT} \sum_{t=1}^T \sum_{i=1}^n X_{it1} \alpha_i \\ \vdots \\ \frac{1}{nT} \sum_{t=1}^T \sum_{i=1}^n X_{itk} \alpha_i \end{bmatrix}$$

$$\text{So plim } \frac{1}{NT} X'Z = \begin{bmatrix} E[X_{i1} Z_i] \\ \vdots \\ E[X_{iT} Z_i] \end{bmatrix} = E[X_{it} Z_i]$$

(b)

$\hat{\beta}^{\text{POLS}}$ is consistent if $E[X_{it} Z_i] = 0$ and otherwise inconsistent

Pooled OLS (POLS) fails when time-invariant unobserved individual characteristics correlate with the regressor and affect the outcome variable

In this case, we can use the panel dimension to control for this.

Question II: Instrumental Variables and Returns to Schooling

This question replicates the instrumental variables analysis in Card (1995), following Cunningham's *Causal Inference: The Mixtape* (Chapter 7, Table 57, p. 355). Card (1995) studies the returns to schooling using **proximity to a four-year college** (`nearc4`) as an instrument for completed years of education (`educ`), with log wages (`lwage`) as the outcome. The identifying idea is that individuals who grew up near a college face a lower cost of attending, inducing variation in educational attainment that is plausibly unrelated to unobserved determinants of wages.

(a) (10 points) Use the Frisch–Waugh–Lovell (FWL) theorem to partial out the included controls and visualize the identifying variation in the data.

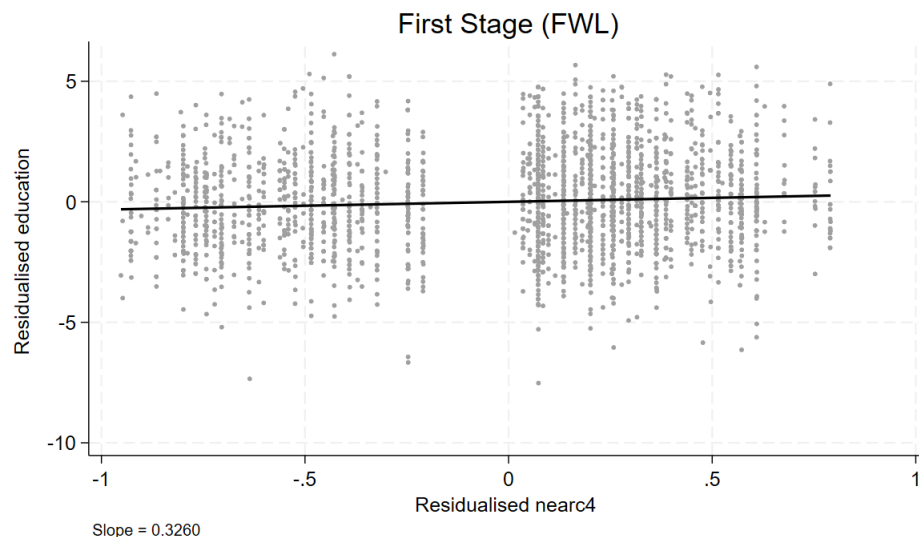
- (i) Let $\mathbf{X}_c$ denote the matrix of included controls (all regressors except `educ` and `nearc4`: experience, its square, `black`, `south`, `smsa`, and region dummies, plus a constant). Regress each of `educ`, `nearc4`, and `lwage` separately on $\mathbf{X}_c$ and save the residuals $\tilde{d}$, $\tilde{z}$, and $\tilde{y}$, respectively. By FWL, these residuals represent the variation in each variable that is orthogonal to the included controls.

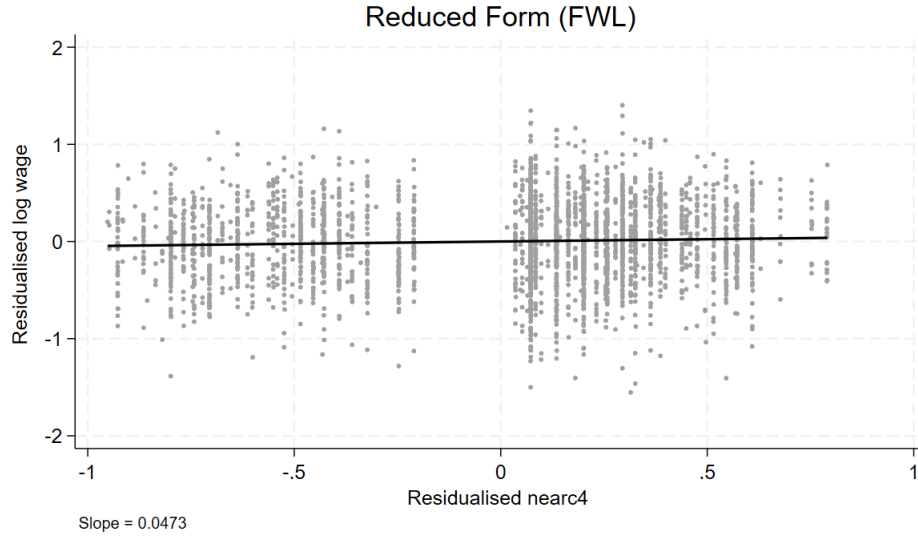
Answer: See Log File

(ii) Produce two scatter plots, each with a fitted regression line overlaid:

- **First stage:** $\tilde{d}$ (residualized education) on the vertical axis against $\tilde{z}$ (residualized `nearc4`) on the horizontal axis.
- **Reduced form:** $\tilde{y}$ (residualized log wage) on the vertical axis against $\tilde{z}$ on the horizontal axis.

Label your axes clearly and include the slope coefficient of each fitted line on the plot or in the caption.





(iii) Using the slope coefficients from (ii), compute the Wald IV estimate as:

$$\hat{\beta}_{Wald}^{IV} = \frac{\text{slope of reduced form}}{\text{slope of first stage}} = \frac{\tilde{z}'\tilde{y}}{\tilde{z}'\tilde{d}} \quad (1)$$

Report your estimate

Answer: Using the Card (1995) data, the Wald estimate is approximately $\hat{\beta}_{Wald}^{IV} \approx 0.145$, implying that an additional year of education induced by college proximity raises log wages by roughly 14.5%.

(iv) Using the first-stage regression from (i), report the F -statistic on `nearc4` and comment on whether the instrument appears **weak**.

Answer: The F -statistic on `nearc4` is approximately $F \approx 15$, exceeding the conventional threshold of 10.

(b) (10 points) **2SLS estimation (replication of Table 57, p. 355).** Replicate the OLS and 2SLS results reported in Table 57 of *The Mixtape*. Interpret results and compare your 2SLS estimate to the IV-Wald estimate you obtained in part (a)

Table 1: Replication Table 57

	OLS	2SLS
educ	0.0712*** (0.0036)	0.1242** (0.0492)
exper	0.0342*** (0.0023)	0.0556*** (0.0199)
black	-0.1660*** (0.0174)	-0.1157** (0.0496)
south	-0.1316*** (0.0152)	-0.1132*** (0.0229)
married	-0.0359*** (0.0036)	-0.0320*** (0.0051)
smsa	0.1758*** (0.0151)	0.1477*** (0.0303)
F-test		17
Observations	3003	3003
R^2	0.305	0.251

Notes: Heteroscedasticity-robust standard errors in parentheses. Controls: region dummies, and a constant. Instrument in 2SLS: `nearc4` (proximity to 4-year college). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

- (c) (8 points) Describe the conditions under which `nearc4` would *fail* to be a valid instrument in this setting. Provide at least two concrete, economically plausible channels through which college proximity could directly affect wages independently of years of schooling completed. For each channel, explain the direction of the resulting bias in the 2SLS estimate.

Answer: For `nearc4` to be a valid instrument for `educ` in the wage equation, it must satisfy:

- **Exclusion restriction:** $\text{Cov}(\text{nearc4}_i, \text{educ}_i) \neq 0$
- **Strong instrument:** `nearc4i` must affect the decision to attend college

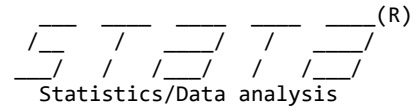
Channel 1: Areas that have four-year colleges tend to be larger, more urbanized labor markets with a denser set of employers, higher demand for skilled workers, and higher average wages across all education levels.

Channel 2: A child raised near a college town may benefit from higher neighborhood quality, better primary and secondary schools, that translates into higher wages through channels other than completed years of schooling.

In both cases the direction of bias is the same: the instrument is positively correlated with the wage residual u_i , so $\hat{\beta}^{2SLS}$ overstates the causal return to schooling.

- (d) (8 points) In words (no algebra required), explain what the 2SLS estimator identifies when the instrument is valid but *heterogeneous treatment effects* are present across individuals.

Answer: When the treatment effect of education on wages varies across individuals the 2SLS estimator identifies the Local Average Treatment Effect (LATE). This is the effect on Compliers, who attend college *only because* they grew up near one



```

    exper |   -.412537   .0337729  -12.22   0.000   -.4787575   -.3463166
    name:  <unnamed>
    log:   C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026\Prob
> lem Set 3\Q2_log.smcl
    log type: smcl
    opened on: 15 Apr 2026, 16:53:17

```

```

1 .
2 . global mydirec="C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026
> \Problem Set 3"

3 .
4 . cd "$mydirec"
   C:\Users\freylejer\OneDrive - UNBC\AP_UNBC\ECON712_Winter_2026\Problem Set 3

5 .
6 . use "card.dta"

7 .
8 . *** PART I, i ***
9 .
10 . reg educ exper expersq black south smsa reg662 reg663 reg664 ///
>      reg665 reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	10230.9465	13	786.995882	F(13, 2996)	=	208.09
Residual	11331.1336	2,996	3.78208732	Prob > F	=	0.0000
				R-squared	=	0.4745
				Adj R-squared	=	0.4722
Total	21562.0801	3,009	7.16586243	Root MSE	=	1.9448

educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	-.412537	.0337729	-12.22	0.000	-.4787575	-.3463166
expersq	.0008736	.0016538	0.53	0.597	-.0023691	.0041163
black	-.9246025	.0937567	-9.86	0.000	-1.108436	-.7407685
south	-.0421616	.1352071	-0.31	0.755	-.3072698	.2229465
smsa	.5188739	.0824334	6.29	0.000	.3572421	.6805057
reg662	-.0742454	.1875067	-0.40	0.692	-.4419004	.2934097
reg663	-.0652009	.1834829	-0.36	0.722	-.4249662	.2945643
reg664	.0651931	.216739	0.30	0.764	-.3597792	.4901653
reg665	-.3522132	.2172502	-1.62	0.105	-.7781878	.0737615
reg666	-.4438211	.2333009	-1.90	0.057	-.9012672	.013625
reg667	-.3170944	.2332312	-1.36	0.174	-.7744039	.1402152
reg668	.4697064	.2676291	1.76	0.079	-.055049	.9944617
reg669	.1951717	.2027681	0.96	0.336	-.202407	.5927504
_cons	16.83694	.2355166	71.49	0.000	16.37515	17.29873

```

11 . predict d_tilde, resid
12 .

```

```

13 . * Residualise nearc4 on X_c → z_tilde
14 . reg nearc4 exper expersq black south smsa reg662 reg663 reg664 reg665 ///
> reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	121.405116	13	9.33885506	F(13, 2996)	=	52.66
Residual	531.326113	2,996	.177345165	Prob > F	=	0.0000
				R-squared	=	0.1860
				Adj R-squared	=	0.1825
Total	652.731229	3,009	.216926298	Root MSE	=	.42112

nearc4	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	-.0000383	.0073133	-0.01	0.996	-.0143779	.0143013
expersq	-2.15e-06	.0003581	-0.01	0.995	-.0007043	.0007
black	.037158	.0203023	1.83	0.067	-.00265	.0769659
south	.0203676	.0292781	0.70	0.487	-.0370396	.0777749
smsa	.3141448	.0178504	17.60	0.000	.2791446	.349145
reg662	.0118388	.0406033	0.29	0.771	-.0677742	.0914519
reg663	-.1166319	.0397319	-2.94	0.003	-.1945365	-.0387273
reg664	-.1472816	.0469333	-3.14	0.002	-.2393063	-.0552569
reg665	-.230613	.047044	-4.90	0.000	-.3228547	-.1383712
reg666	-.4115685	.0505196	-8.15	0.000	-.5106252	-.3125118
reg667	-.2987625	.0505046	-5.92	0.000	-.3977896	-.1997354
reg668	-.1698296	.0579532	-2.93	0.003	-.2834616	-.0561975
reg669	-.0506329	.043908	-1.15	0.249	-.1367258	.0354599
_cons	.6018745	.0509994	11.80	0.000	.5018771	.701872

```

15 . predict z_tilde, resid

```

```

16 .
17 . * Residualise lwage on X_c → y_tilde
18 .
19 . reg lwage exper expersq black south smsa reg662 reg663 reg664 reg665 ///
> reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	114.030686	13	8.77159122	F(13, 2996)	=	54.91
Residual	478.610925	2,996	.159749975	Prob > F	=	0.0000
				R-squared	=	0.1924
				Adj R-squared	=	0.1889
Total	592.641611	3,009	.196956335	Root MSE	=	.39969

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
exper	.0540272	.006941	7.78	0.000	.0404175	.0676368
expersq	-.0022159	.0003399	-6.52	0.000	-.0028823	-.0015494
black	-.2690631	.0192689	-13.96	0.000	-.3068447	-.2312815
south	-.1480811	.0277878	-5.33	0.000	-.2025662	-.093596
smsa	.1918217	.0169417	11.32	0.000	.1586031	.2250403
reg662	.0914433	.0385364	2.37	0.018	.0158827	.1670038
reg663	.1397198	.0377095	3.71	0.000	.0657808	.2136589
reg664	.0549257	.0445442	1.23	0.218	-.0324147	.1422662
reg665	.095719	.0446493	2.14	0.032	.0081726	.1832654
reg666	.0977417	.0479481	2.04	0.042	.0037273	.1917562
reg667	.0894529	.0479337	1.87	0.062	-.0045335	.1834393
reg668	-.0212046	.0550032	-0.39	0.700	-.1290525	.0866432
reg669	.1342786	.0416729	3.22	0.001	.0525682	.2159891
_cons	5.884744	.0484034	121.58	0.000	5.789837	5.979652

20 . predict y_tilde, resid

21 .

22 .

23 . *** PART I, ii ***

24 .

25 . reg d_tilde z_tilde

Source	SS	df	MS	Number of obs	=	3,010
Model	56.4529005	1	56.4529005	F(1, 3008)	=	15.06
Residual	11274.6807	3,008	3.74823161	Prob > F	=	0.0001
				R-squared	=	0.0050
				Adj R-squared	=	0.0047
Total	11331.1336	3,009	3.76574729	Root MSE	=	1.936

d_tilde	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
z_tilde	.3259587	.083991	3.88	0.000	.1612731	.4906442
_cons	2.19e-09	.0352882	0.00	1.000	-.0691915	.0691915

26 . local b_fs = string(_b[z_tilde], "%6.4f")

27 .

28 . reg y_tilde z_tilde

Source	SS	df	MS	Number of obs	=	3,010
Model	1.18731521	1	1.18731521	F(1, 3008)	=	7.48
Residual	477.423609	3,008	.158717955	Prob > F	=	0.0063
				R-squared	=	0.0025
				Adj R-squared	=	0.0021
Total	478.610924	3,009	.159059795	Root MSE	=	.39839

y_tilde	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
z_tilde	.0472718	.0172835	2.74	0.006	.0133831	.0811606
_cons	1.49e-11	.0072616	0.00	1.000	-.0142381	.0142381

29 . local b_rf = string(_b[z_tilde], "%6.4f")

30 .

```
31 . twoway (scatter d_tilde z_tilde, msize(tiny) mcolor(gs10)) ///
>         (lfit d_tilde z_tilde, lcolor(black) lwidth(medthick)), ///
>         xtitle("Residualised nearc4 ") ///
>         ytitle("Residualised education ") ///
>         title("First Stage (FWL)") ///
>         note("Slope = `b_fs'", size(small)) ///
>         legend(off)
```

32 . quietly graph export "PS3_Q2a_FirstStage.png", replace width(1200)

```

33 .
34 . * Reduced-form scatter: y_tilde on z_tilde
35 . twoway (scatter y_tilde z_tilde, msize(tiny) mcolor(gs10)) ///
> (lfit y_tilde z_tilde, lcolor(black) lwidth(medthick)), ///
> xtitle("Residualised nearc4 ") ///
> ytitle("Residualised log wage ") ///
> title("Reduced Form (FWL)") ///
> note("Slope = `b_rf'", size(small)) ///
> legend(off)

36 . quietly graph export "PS3_Q2a_ReducedForm.png", replace width(1200)

37 .
38 .
39 . *** PART I, iii ***
40 .
41 . reg educ nearc4 exper expersq black south smsa reg662 reg663 reg664 reg665 ///
> reg666 reg667 reg668 reg669

```

Source	SS	df	MS	Number of obs	=	3,010
Model	10287.3994	14	734.81424	F(14, 2995)	=	195.20
Residual	11274.6807	2,995	3.76450107	Prob > F	=	0.0000
				R-squared	=	0.4771
				Adj R-squared	=	0.4747
Total	21562.0801	3,009	7.16586243	Root MSE	=	1.9402

educ	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
nearc4	.3259587	.0841731	3.87	0.000	.1609158	.4910015
exper	-.4125245	.0336943	-12.24	0.000	-.4785909	-.3464582
expersq	.0008743	.0016499	0.53	0.596	-.0023608	.0041095
black	-.9367145	.0935907	-10.01	0.000	-1.120223	-.7532059
south	-.0488006	.1349033	-0.36	0.718	-.3133131	.2157118
smsa	.4164757	.0863879	4.82	0.000	.24709	.5858614
reg662	-.0781043	.187073	-0.42	0.676	-.4449088	.2887002
reg663	-.0271837	.1833189	-0.15	0.882	-.3866274	.33226
reg664	.1132008	.2165896	0.52	0.601	-.3114786	.5378801
reg665	-.2770429	.217612	-1.27	0.203	-.703727	.1496413
reg666	-.3096667	.2353218	-1.32	0.188	-.7710754	.1517419
reg667	-.2197101	.2340433	-0.94	0.348	-.6786121	.2391918
reg668	.5250638	.2673885	1.96	0.050	.00078	1.049348
reg669	.2116759	.202341	1.05	0.296	-.1850654	.6084173
_cons	16.64075	.2403679	69.23	0.000	16.16945	17.11206

```

42 . test nearc4

```

(1) nearc4 = 0

F(1, 2995) = 15.00
Prob > F = 0.0001

```
43 . display "F-stat on nearc4 = " %6.2f r(F) " (p = " %5.3f r(p) ")"
    F-stat on nearc4 = 15.00 (p = 0.000)
```

```
44 .
45 .
46 . ***** PART II *****
47 .
48 .
49 . reg lwage educ exper black south married smsa, robust
```

```
Linear regression               Number of obs   =    3,003
                               F(6, 2996)       =   230.88
                               Prob > F         =    0.0000
                               R-squared        =    0.3050
                               Root MSE     =    .37025
```

lwage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
educ	.0711729	.0036137	19.70	0.000	.0640872	.0782585
exper	.0341518	.0022668	15.07	0.000	.0297072	.0385964
black	-.1660274	.0173977	-9.54	0.000	-.2001402	-.1319147
south	-.1315518	.0152007	-8.65	0.000	-.1613567	-.1017469
married	-.0358707	.0035727	-10.04	0.000	-.042876	-.0288655
smsa	.1757871	.0150994	11.64	0.000	.1461808	.2053934
_cons	5.063317	.0661786	76.51	0.000	4.933557	5.193077

```
50 . eststo OLS
51 .
52 .
53 . ivreg2 lwage (educ = nearc4) exper black south married smsa, first ffirst ro
    > bust
```

First-stage regressions

First-stage regression of educ:

Statistics robust to heteroskedasticity
Number of obs = 3003

educ	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
nearc4	.3272826	.0805341	4.06	0.000	.1693748	.4851903
exper	-.404434	.0081984	-49.33	0.000	-.420509	-.388359
black	-.9475281	.0884964	-10.71	0.000	-1.121048	-.7740082
south	-.2973528	.0787506	-3.78	0.000	-.4517635	-.1429421
married	-.0726936	.0176169	-4.13	0.000	-.107236	-.0381512
smsa	.4208945	.0848857	4.96	0.000	.2544543	.5873348
_cons	16.8307	.1252943	134.33	0.000	16.58502	17.07637

F test of excluded instruments:

F(1, 2996) = 16.52
Prob > F = 0.0000

Sanderson-Windmeijer multivariate F test of excluded instruments:

F(1, 2996) = 16.52
Prob > F = 0.0000

Summary results for first-stage regressions

Variable	F(1, 2996)	P-val	(Underid)		P-val	(Weak id)	
			SW	Chi-sq(1)		SW	F(1, 2996)
educ	16.52	0.0000		16.55	0.0000		16.52

NB: first-stage test statistics heteroskedasticity-robust

Stock-Yogo weak ID F test critical values for single endogenous regressor:

10% maximal IV size	16.38
15% maximal IV size	8.96
20% maximal IV size	6.66
25% maximal IV size	5.53

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for i.i.d. errors only.

Underidentification test

Ho: matrix of reduced form coefficients has rank=K1-1 (underidentified)

Ha: matrix has rank=K1 (identified)

Kleibergen-Paap rk LM statistic Chi-sq(1)=16.36 P-val=0.0001

Weak identification test

Ho: equation is weakly identified

Cragg-Donald Wald F statistic 15.77

Kleibergen-Paap Wald rk F statistic 16.52

Stock-Yogo weak ID test critical values for K1=1 and L1=1:

10% maximal IV size	16.38
15% maximal IV size	8.96
20% maximal IV size	6.66
25% maximal IV size	5.53

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.

Weak-instrument-robust inference

Tests of joint significance of endogenous regressors B1 in main equation

Ho: B1=0 and orthogonality conditions are valid

Anderson-Rubin Wald test F(1,2996)= 6.34 P-val=0.0119

Anderson-Rubin Wald test Chi-sq(1)= 6.35 P-val=0.0117

Stock-Wright LM S statistic Chi-sq(1)= 6.36 P-val=0.0117

NB: Underidentification, weak identification and weak-identification-robust test statistics heteroskedasticity-robust

Number of observations	N =	3003
Number of regressors	K =	7
Number of endogenous regressors	K1 =	1
Number of instruments	L =	7
Number of excluded instruments	L1 =	1

IV (2SLS) estimation

Estimates efficient for homoskedasticity only

Statistics robust to heteroskedasticity

		Number of obs =	3003
		F(6, 2996) =	148.44
		Prob > F =	0.0000
Total (centered) SS	=	Centered R2 =	0.2513
Total (uncentered) SS	=	Uncentered R2 =	0.9963
Residual SS	=	Root MSE =	.3838

lwage	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
educ	.1241642	.0491577	2.53	0.012	.0278169	.2205116
exper	.0555882	.0198674	2.80	0.005	.0166488	.0945277
black	-.1156855	.0495591	-2.33	0.020	-.2128197	-.0185514
south	-.1131647	.0229476	-4.93	0.000	-.1581411	-.0681883
married	-.0319754	.0050725	-6.30	0.000	-.0419172	-.0220335
smsa	.1477065	.0303171	4.87	0.000	.0882861	.2071269
_cons	4.162476	.8348751	4.99	0.000	2.526151	5.798801

Underidentification test (Kleibergen-Paap rk LM statistic): **16.356**
Chi-sq(1) P-val = **0.0001**

Weak identification test (Cragg-Donald Wald F statistic): **15.767**
(Kleibergen-Paap rk Wald F statistic): **16.515**
Stock-Yogo weak ID test critical values: 10% maximal IV size **16.38**
15% maximal IV size **8.96**
20% maximal IV size **6.66**
25% maximal IV size **5.53**

Source: Stock-Yogo (2005). Reproduced by permission.

NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.

Hansen J statistic (overidentification test of all instruments): **0.000**
(equation exactly identified)

Instrumented: educ
Included instruments: exper black south married smsa
Excluded instruments: nearc4

54 . eststo tsls

55 . matrix first=e(first)

56 . estadd scalar F1=first[4,1]

added scalar:

$$e(F1) = 16.515286$$

57 .

58 .

59 .

```
60 . estout OLS tsls using "PS3_Q2b_Card_Table57.tex",          ///
>   replace                                                    ///
>   style(tex)                                                  ///
>   cells(b(star fmt(%9.4f)) se(par fmt(%9.4f)))              ///
>   starlevels(* 0.10 ** 0.05 *** 0.01)                       ///
>   stats(F1 N r2,                                             ///
>     labels("F-test" "Observations" "\$R^2\$"))              //
> /
>   fmt(%9.0f %9.0f %9.3f))                                     ///
>   mlabels("OLS" "2SLS")                                       ///
>   collabels(none)                                             ///
>   keep(educ exper black south married smsa)                  //
> /
>   prehead(                                                    ///
>     "\begin{table}[H]"                                         ///
>     "\begin{center}"                                           ///
>     "\caption{Replication Table 57}"                           ///
>   ///
>     "\resizebox{.57\textwidth}{!}{%"                          ///
>     "\begin{tabular}{lll}"                                     ///
>     "\hline" "\hline"                                         ///
```

```

>      )                                     ///
>      postfoot(                             ///
>      "\hline" "\hline"                     ///
>      "\end{tabular}}"                       ///
>      "\end{center}"                         ///
>      "\captionsetup{font={scriptsize}}"     ///
>      "\captionsetup{width=0.80\textwidth}"  ///
>      "\captionsetup{skip=0pt}"              ///
>      "\caption*{\scriptsize Notes: Heteroscedasticity-robust standard error
> s in parentheses." ///
>      "Controls: region dummies, and a constant." ///
>      "Instrument in 2SLS: \texttt{nearc4} (proximity to 4-year college)." /
> //
>      "* \p<0.10\$, ** \p<0.05\$, *** \p<0.01\$." ///
>      "\end{table}"                          ///
>      )
(output written to PS3_Q2b Card Table57.tex)

```

```

61 .
62 . cap log close

```